ME-203A: Kinematics of Machines

by

Dr. Emarti Kumari

Department of Mechanical Engineering
M.B.M. Engineering College
Jai Narain Vyas University, Jodhpur

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

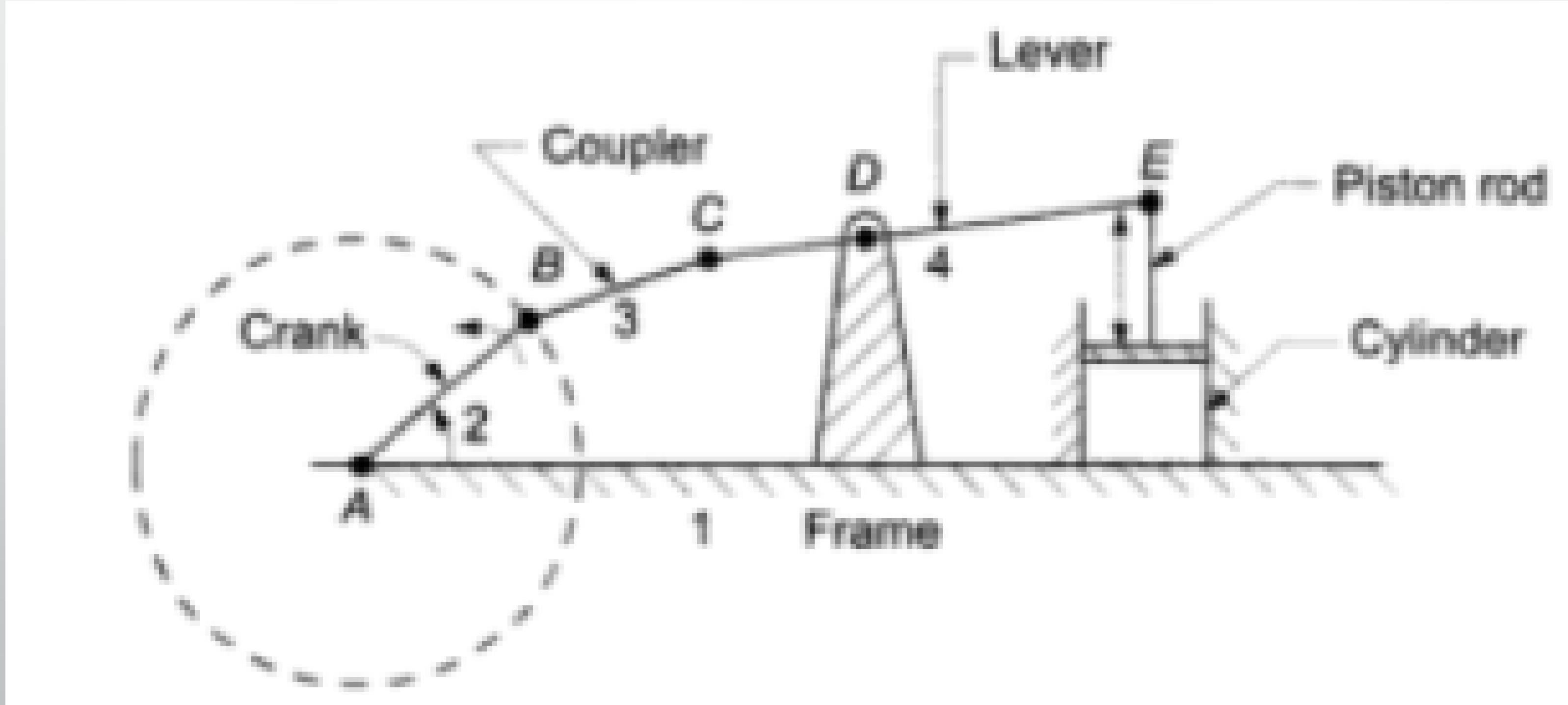
Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

Inversions of Mechanisms:

- A kinematic chain becomes a mechanism when one of the link is fixed.
- Therefore, as many number of mechanisms can be obtained as many are the links in the kinematic chain.
- This method of obtaining different mechanisms by fixing different links of a kinematic chain is called inversion of the mechanism.
- The relative motion between the various links is not altered as a result of inversion, but their absolute motion with respect to the fixed link may alter drastically.

Inversions of a for-bar chain are:

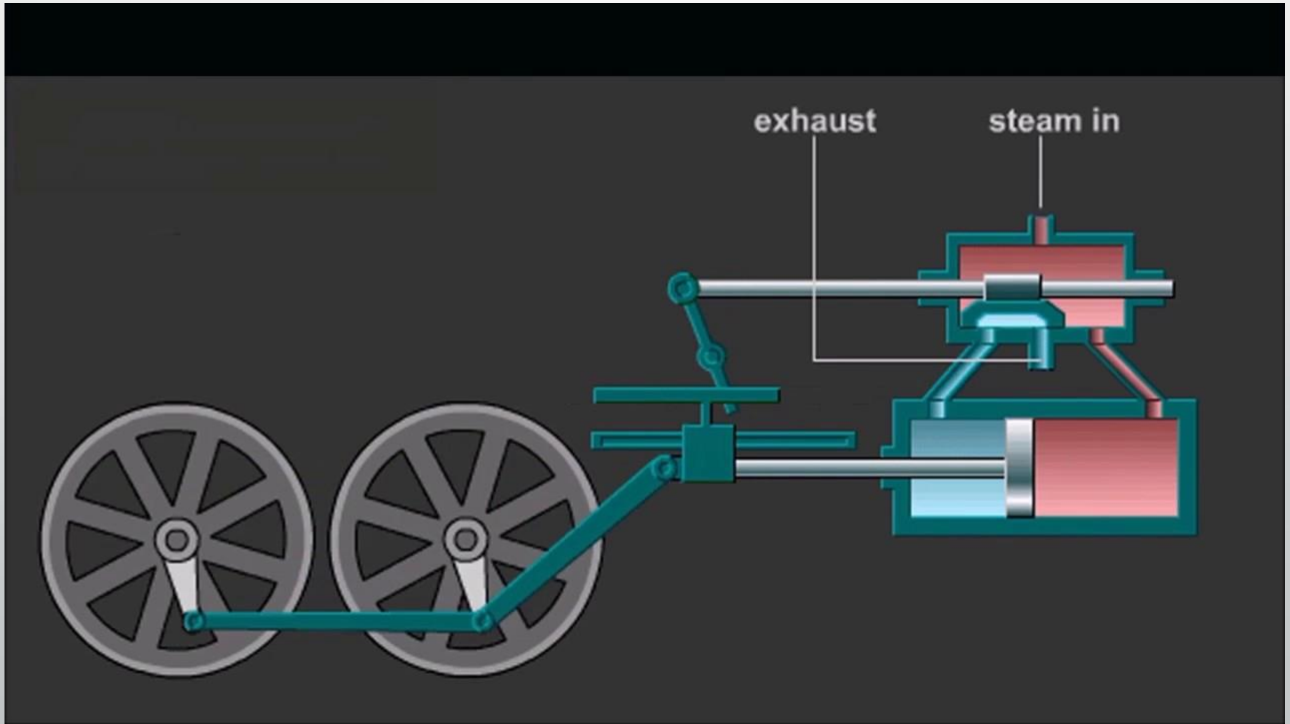
- Beam engine (crank and lever mechanism): a beam engine is a type of steam engine where a pivoted overhead beam is used to apply the force from a vertical piston to a vertical connecting rod.
- Beam engines were first used to pump water out of mines or into canals.



Inversions of a for-bar chain are:

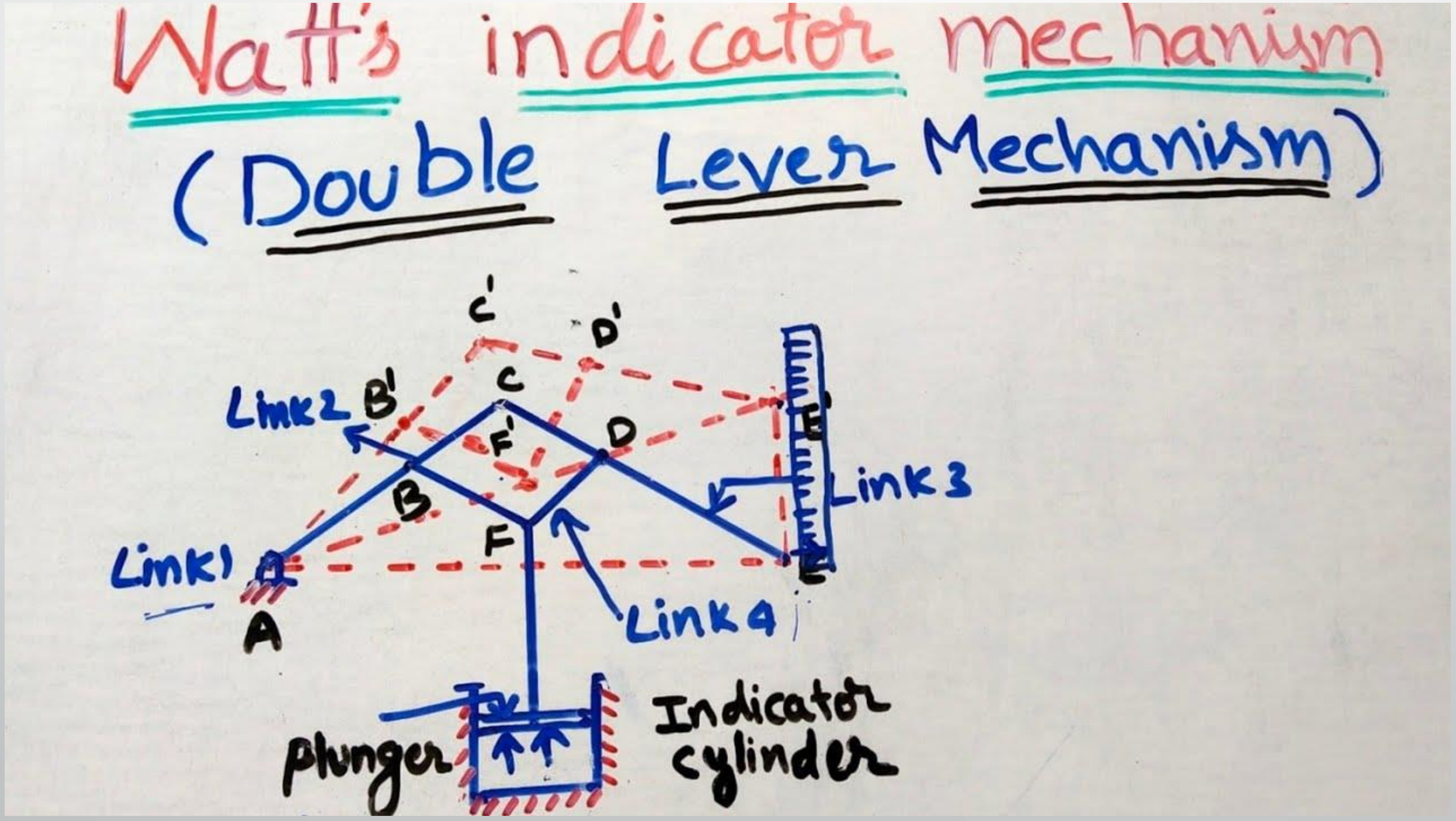
Coupled wheel of locomotive:

- The links AB and CD are of equal lengths and act as crank.
- These cranks are connected to the respective wheels.
- This mechanism is used to transmit rotary motion from one wheel to the other.
- This is also called the double **crank mechanism.**



Inversions of a for-bar chain are:

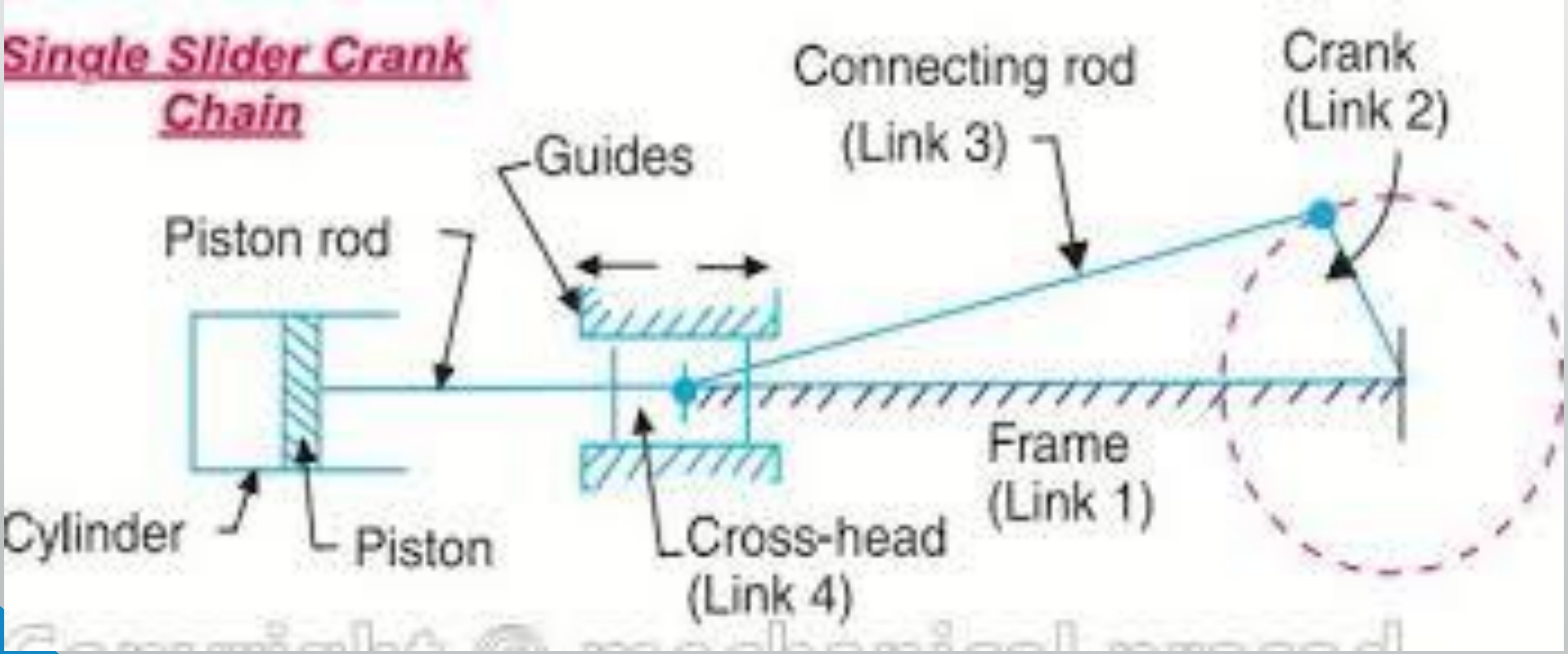
- Watt's indicator mechanism: it consists of four links: a fixed link at A, link AC, link CE, and link BFD.
- **The links CE and BFD acts as levers.**
- The displacement of link BFD is directly proportional to the pressure in the indicator cylinder.



Inversions of a for-bar chain are:

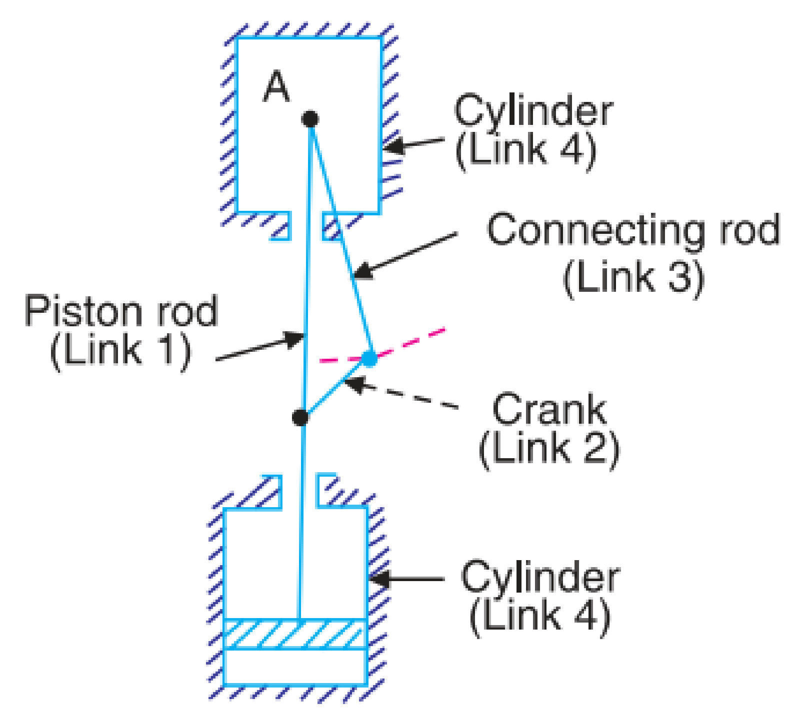
Single slider-crank chain: it consists of three turning pairs and one sliding pair.

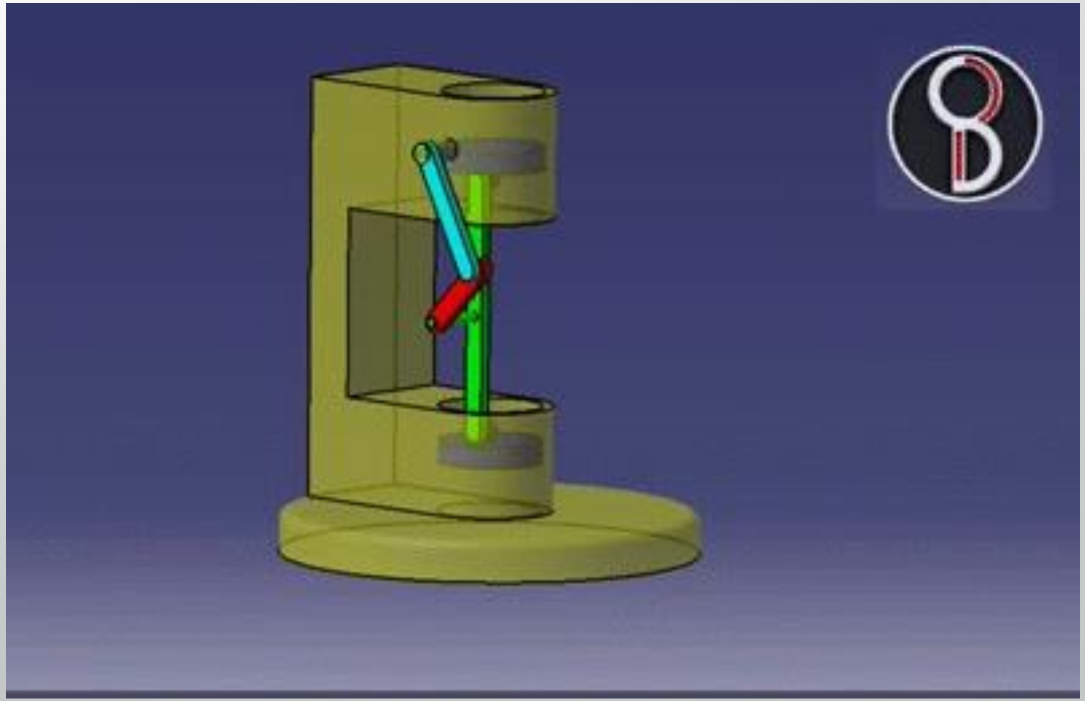
- It is used to convert rotary motion to reciprocating and vice-versa.
- Applications: steam engine, internal combustion engine, reciprocating compressor.



Inversions of a Single-Slider Crank chain are:

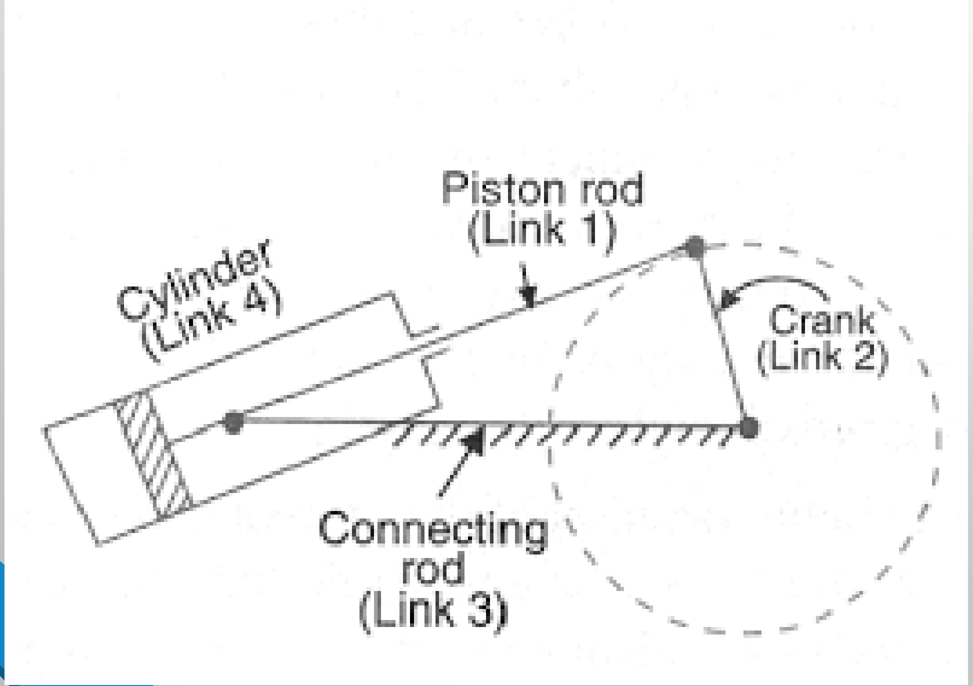
- **Pendulum Pump:** this inversion mechanism is obtained by fixing the link 4.
- When link 2 (crank) rotates, the link 3 (connecting rod) oscillates about the a pin pivoted to fixed link 4 at A and the piston attached to the piston rod (link 1) reciprocates in the cylinder.

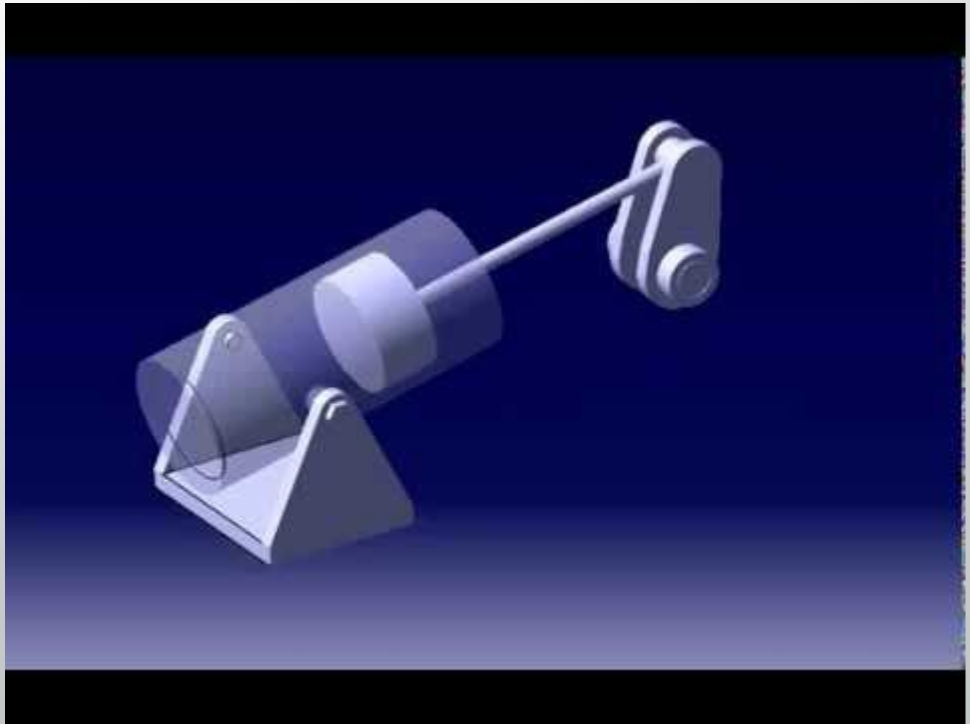




Inversions of a Single-Slider Crank chain are:

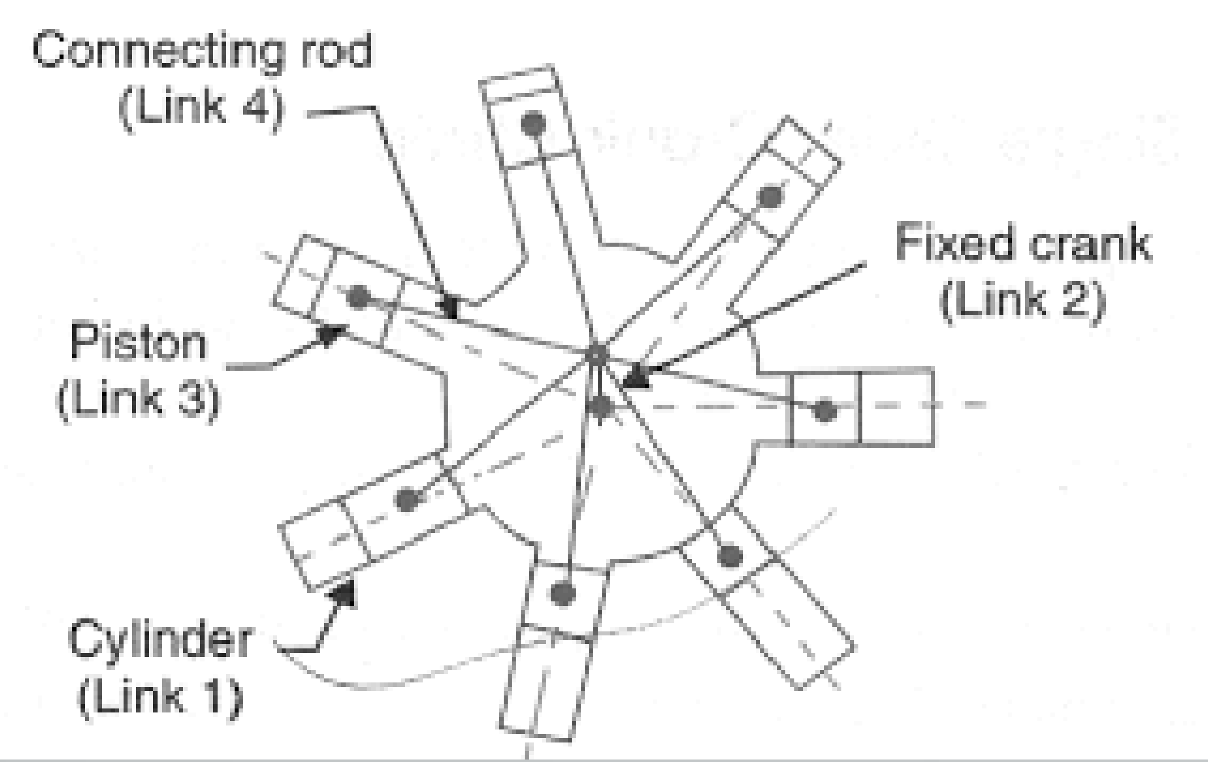
- **Oscillating cylinder engine:** In this mechanism link 3 is fixed.
- When the crank (link 2) rotates, the piston attached to piston rod (link 1) reciprocates and the cylinder (link 4) oscillates about a pin pivoted to the fixed link at A.





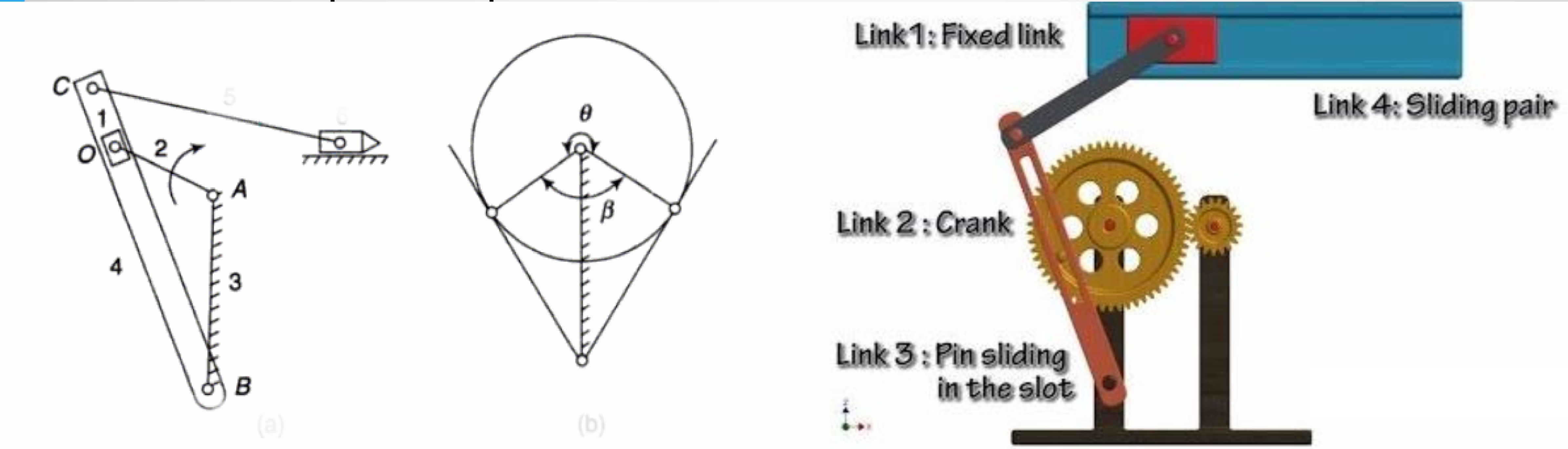
Inversions of a Single-Slider Crank chain are:

- **Rotary internal combustion engine (Gnome engine):** It consists of several cylinder in one plane and all revolve about fixed centre O.
- The crank (link 2) is fixed.
- When the connecting rod (link 4) rotates, the piston (link 3) reciprocates inside the cylinder forming link 1.



Inversions of a Single-Slider Crank chain are:

- **Crank and slotted lever quick-return motion mechanism:** In this mechanism link 3 corresponding to the connecting rod is fixed.
- The driving crank OA (link 2) revolves about centre O.
- A slider (link 1) attached to the crank pin at B slides along the slotted lever CB (link 4) and make the slotted lever oscillate about the pivoted point A.



Inversions of a Single-Slider Crank chain are:

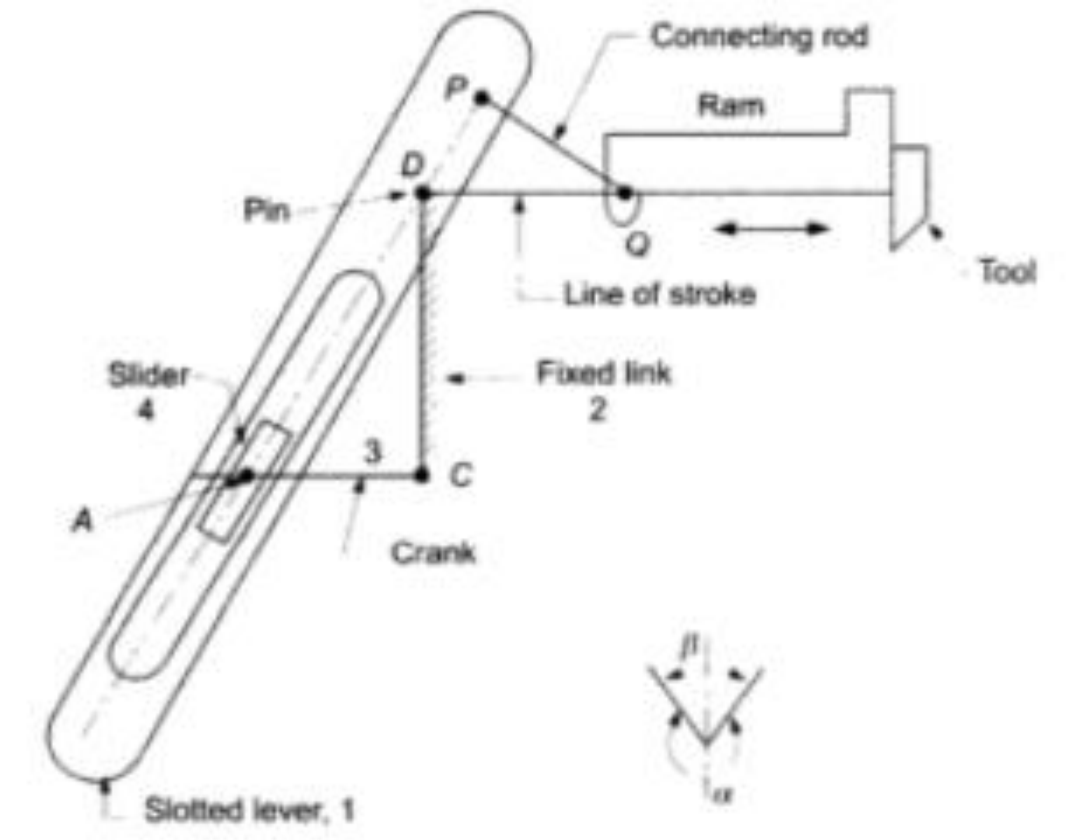
- **Crank and slotted lever quick-return motion mechanism:** In this mechanism link 3 corresponding to the connecting rod is fixed.
- The driving crank OA (link 2) revolves about centre O.
- A slider (link 1) attached to the crank pin at B slides along the slotted lever CB (link 4) and make the slotted lever oscillate about the pivoted point A.
- A short link CD transmits the motion from BC to the arm which reciprocates with the tool along the line of stroke.
- The line of stroke is perpendicular to AB produced.
- This mechanism is mostly used in shaping machines, slotting machine, and rotary internal combustion engines.

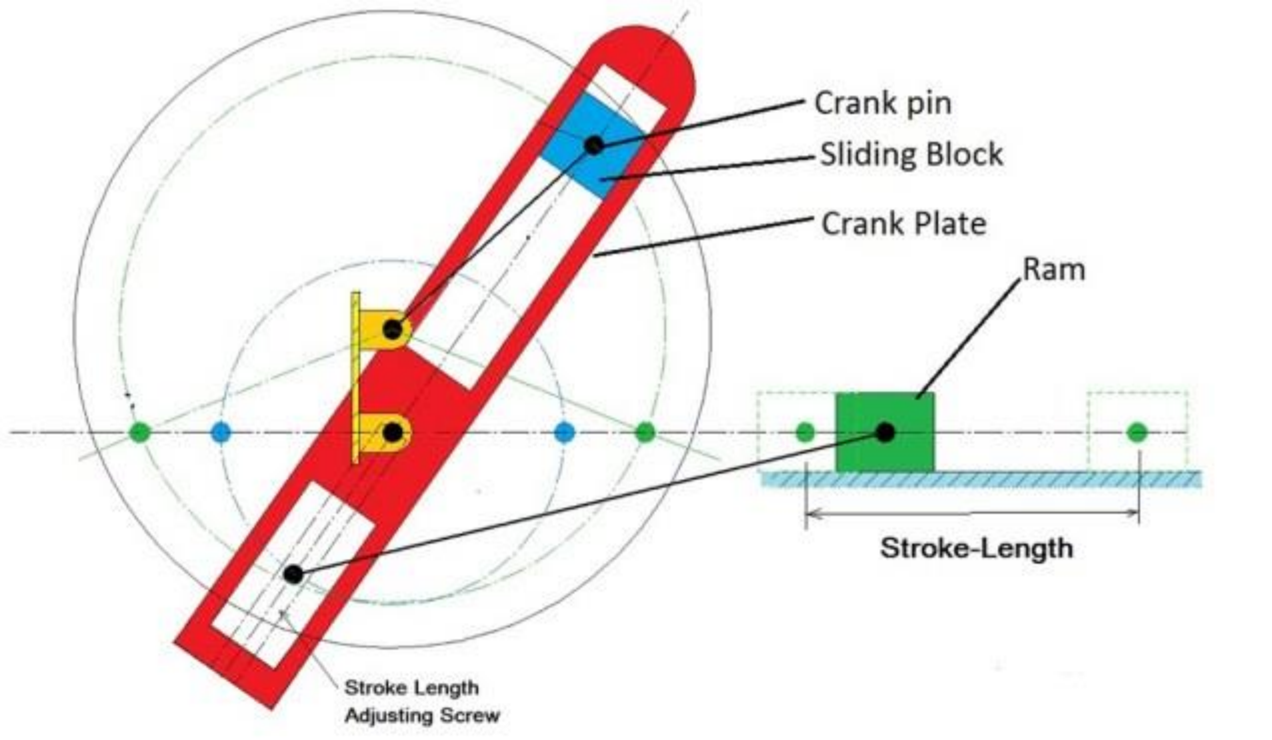
Time of cutting stroke / Time of return stroke = α / β

Length of stroke = 2 BC (AO / AB)

Inversions of a Single-Slider Crank chain are:

- **Whitworth quick-return motion mechanism:** In this mechanism link CD (link 2) is fixed.
- The driving crank CA (link 3) rotates about C.
- The slider (link 4) attached to the pin at A slides along the slotted lever PA (link 1), which oscillates about pivot D.





Withworth quick return mechanism

Inversions of a Single-Slider Crank chain are:

- **Whitworth quick-return motion mechanism:** In this mechanism link CD (link 2) is fixed.
- The driving crank CA (link 3) rotates about C.
- The slider (link 4) attached to the pin at A slides along the slotted lever PA (link 1), which oscillates about pivot D.
- The connecting rod PQ carries the ram at Q with cutting tool.
- The ram reciprocates along the line of stroke,
- It is used in shaping and slotting machines.

Time of cutting stroke / Time of return stroke = α / β

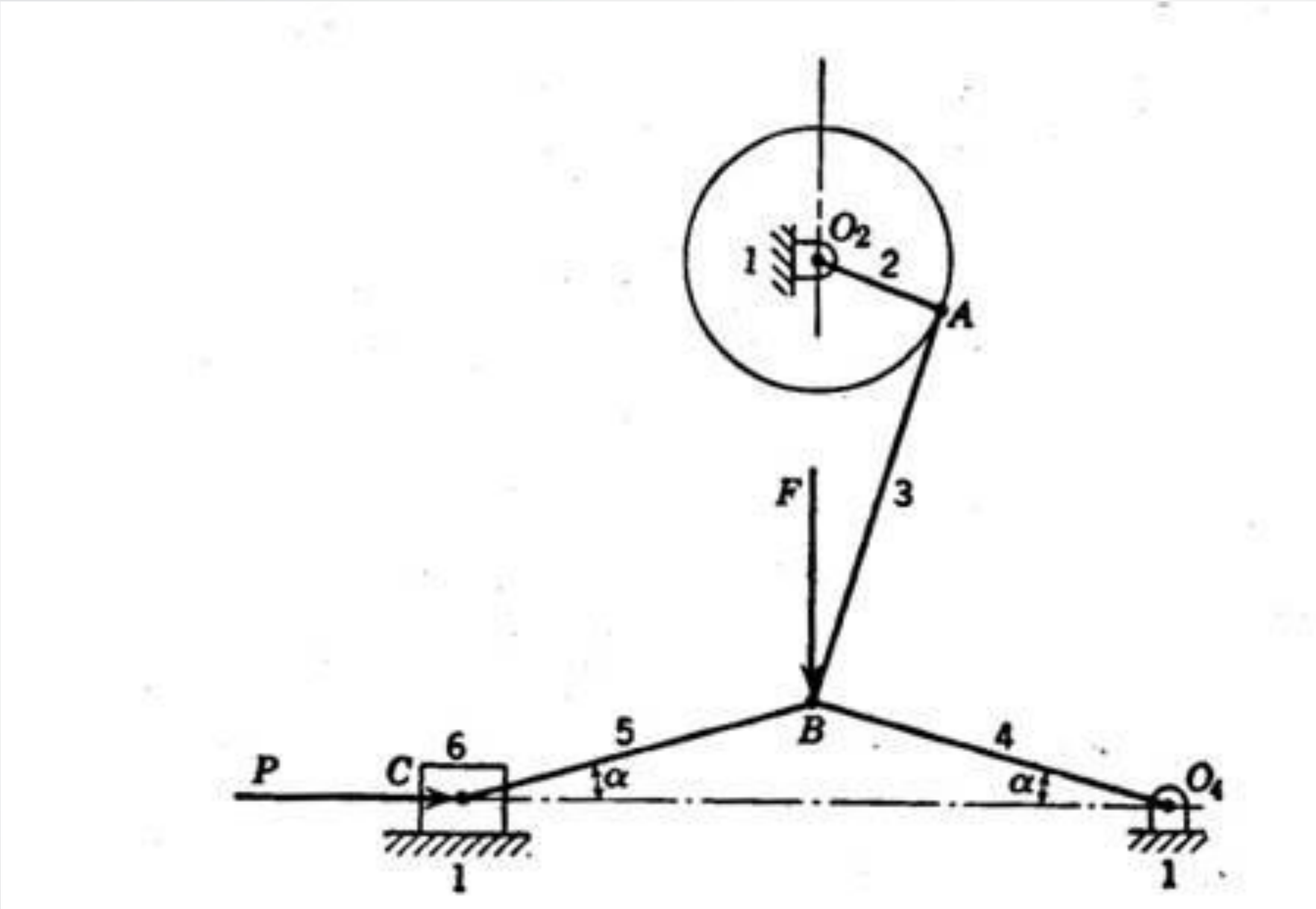
Length of stroke = 2 PD

Inversions of a Single-Slider Crank chain are:

- **Toggle mechanism:** This mechanism has many applications where it is necessary to overcome a large resistance with a small driving force.
- Links 4 and 5 are equal length.
- As the angles α decrease and links 4 and 5 approach being collinear, the force F required to overcome a given resistance P decreases as:

$$F = 2P \tan \alpha$$

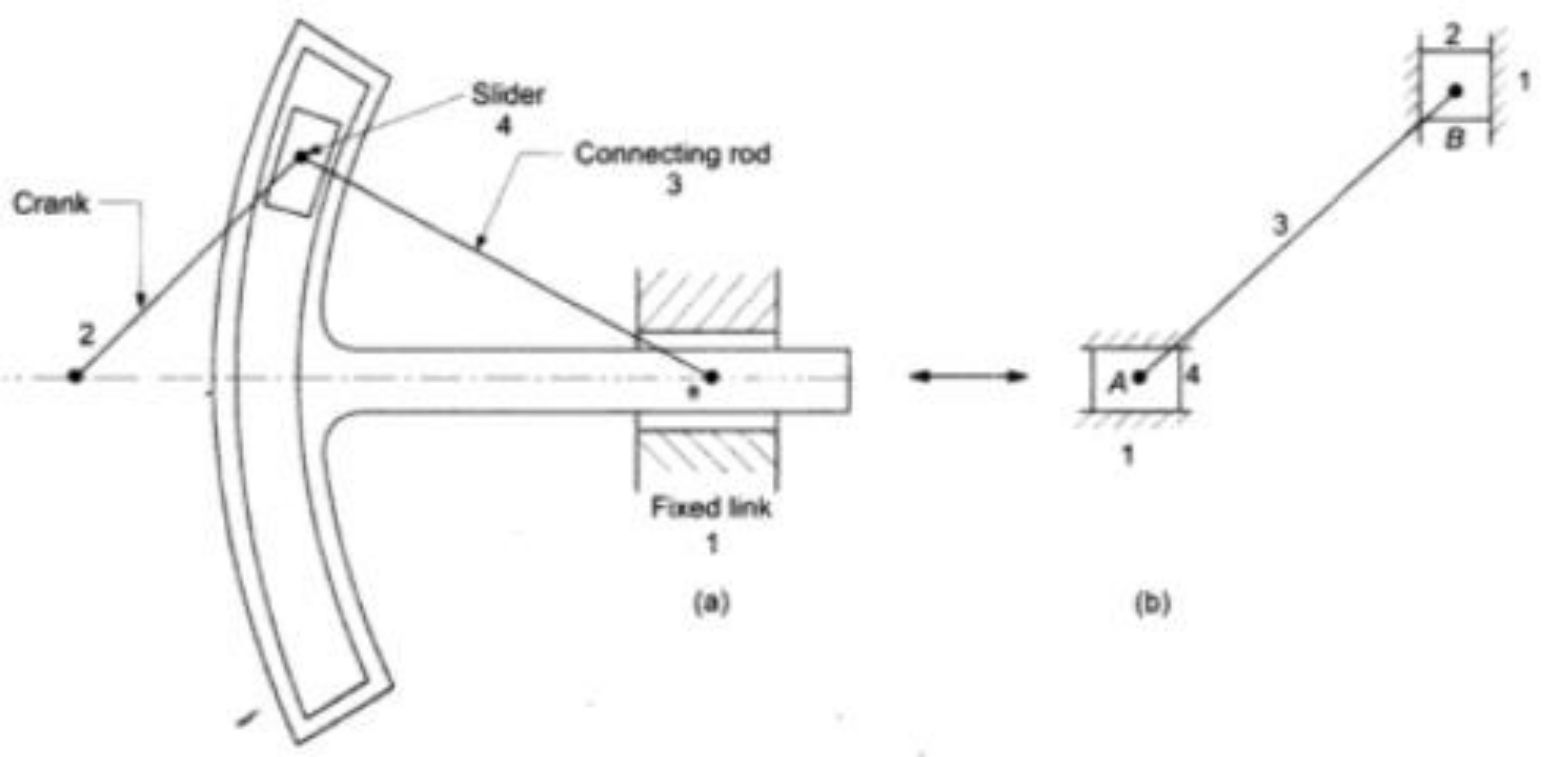
- If α approaches zero, for a given F, P approaches infinity.
- A stone crusher utilizes this mechanism to overcome a large resistance with a small force.
- It can be used in numerous toggle clamping devices, for holding work pieces.



O_2
1
2
A
F
3
B
5
4
6
P
C
α
α
O_4
1
1

Inversions of a Double Slider Crank Chain are:

- A kinematic chain consisting of two turning pairs sliding pairs is called double slider-crank chain.
- Links 3 and 4 reciprocate, link 2 rotates and link 1 is fixed.
- Two pairs of the same kind are adjacent.



Inversions of a Double Slider Crank Chain are:

Donkey Pump:

- Link 2 (crank) rotates about point A.
- One end of the crank is connected to the piston, through the piston rod, which reciprocates vertically in the pump cylinder.
- This cylinder together with the body of the pump represents the fixed link 1.
- The other end of the crank is connected to the slider (link 3) which reciprocates horizontally in the cylinder.

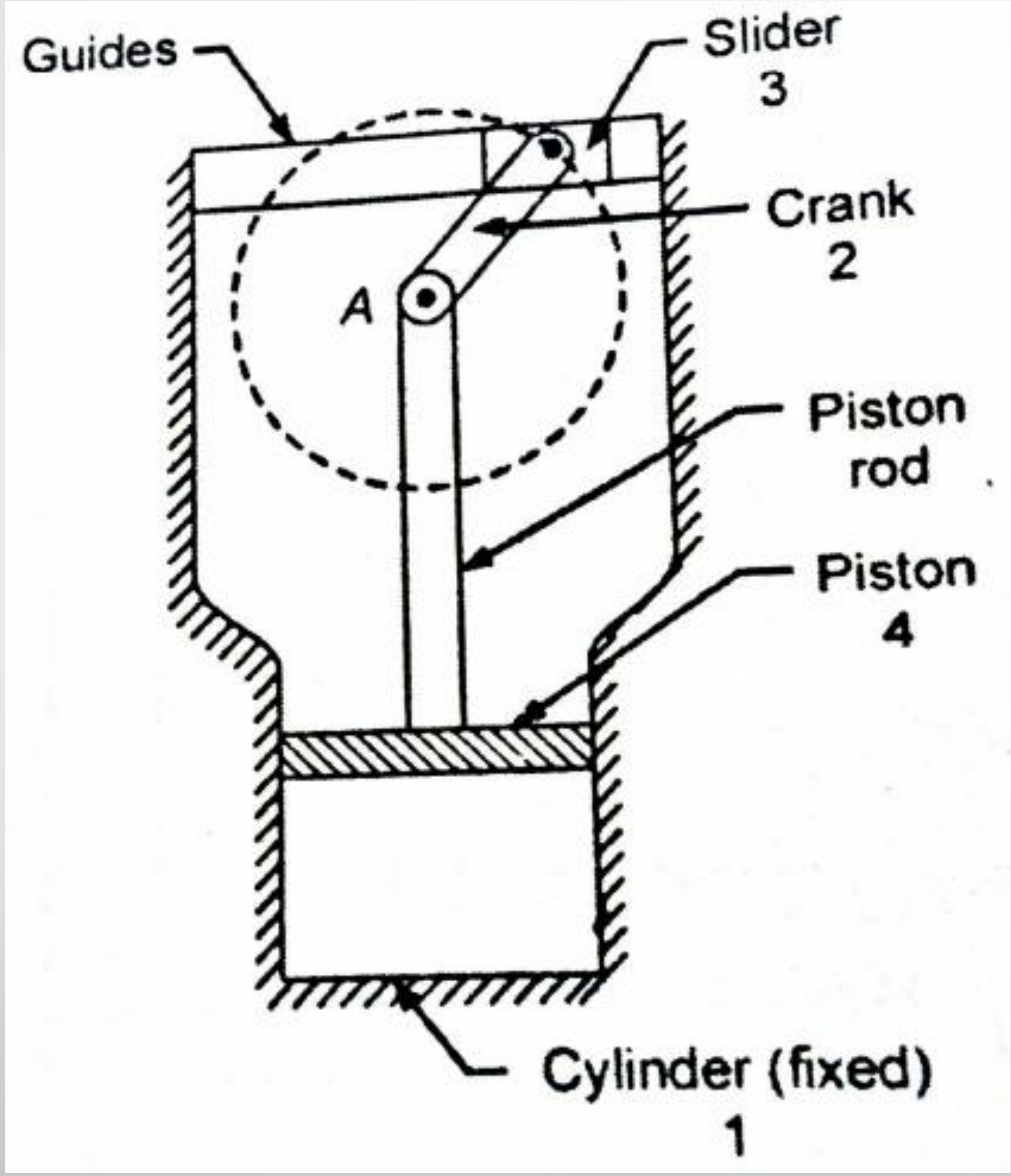
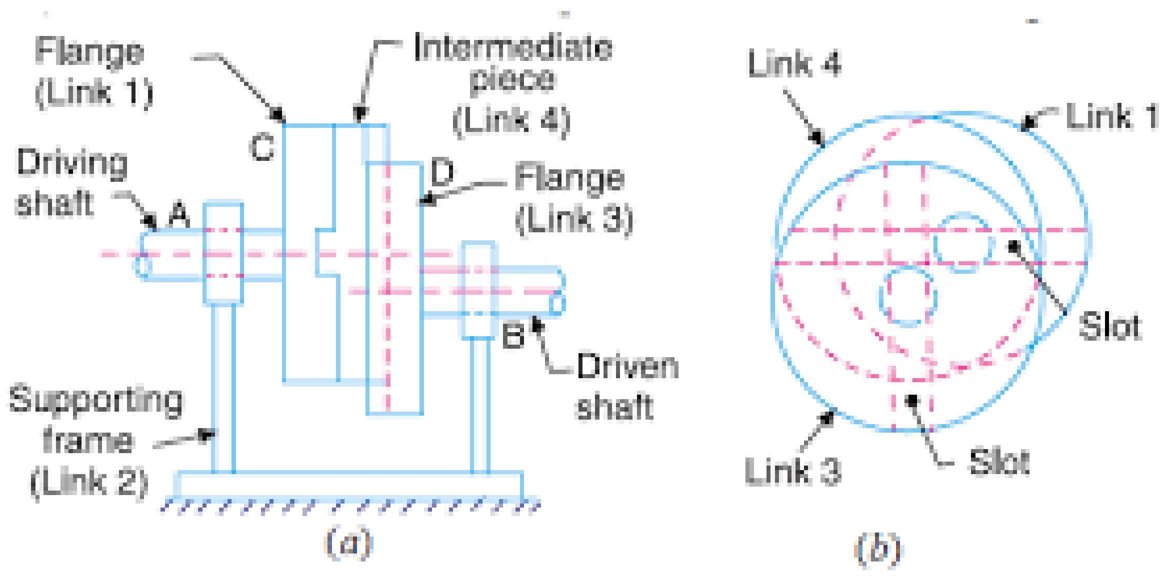


Fig. Donkey Pump

Inversions of a Double Slider Crank Chain are:

Oldham's coupling:

- It is used to connect two parallel shaft, the distance between whose axes is small and variable.
- The shafts connected by the coupling rotate at the same speed.
- The shafts have flange at the ends, in which slots are cut.
- These form a link 1 and 3.



Oldham's coupling.

Inversions of a Double Slider Crank Chain are:

Elliptical trammel:

- It is a device to draw ellipse.
- In which two grooves are cut at right angles in a plate that is fixed.
- The plate forms the fixed link 4.
- Two sliding blocks are fitted into the grooves.
- The sliders form two sliding links 1 and 3.
- The link joining slides form the link 2.

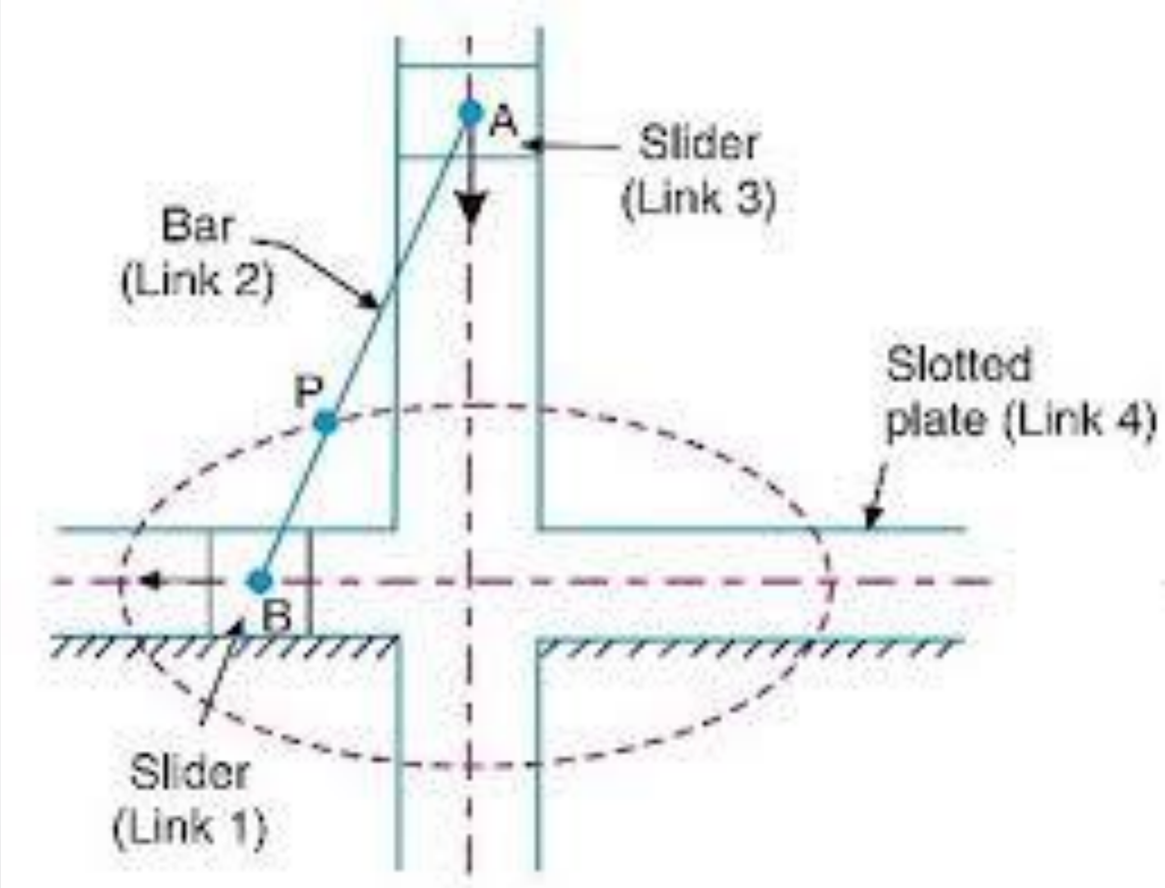
$$y = AB\sin\theta$$

or

$$\frac{y}{AB} = \sin\theta$$

Squaring and adding, we get

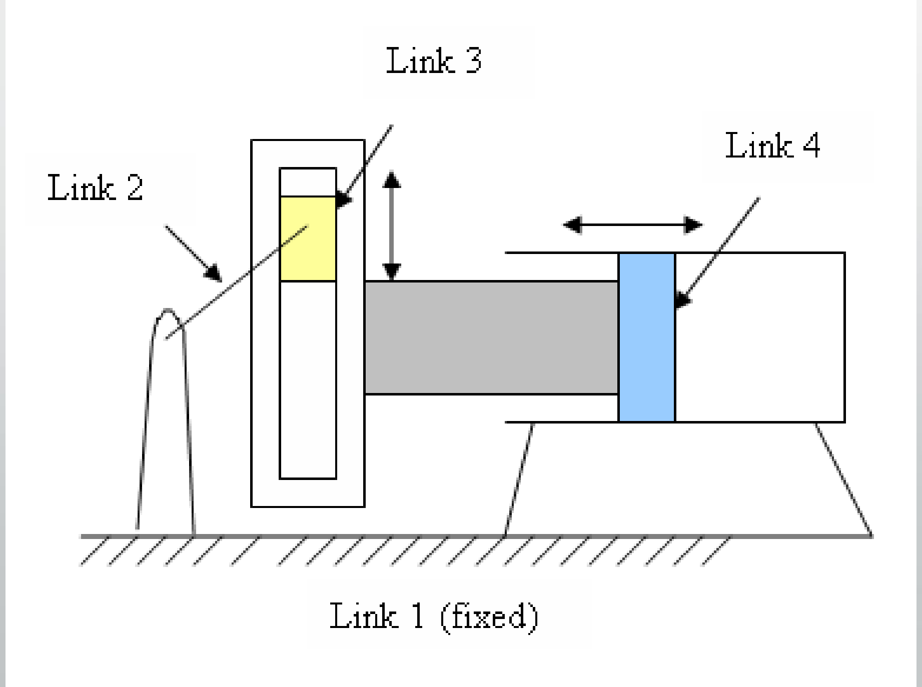
$$\frac{x^2}{BP^2} + \frac{y^2}{AB^2} = 1$$


A
Slider
(Link 3)
Bar
(Link 2)
P
Slotted
plate (Link 4)
B
Slider
(Link 1)

Inversions of a Double Slider Crank Chain are:

Scotch yoke:

- This mechanism gives simple harmonic motion.
- Its early application was on steam pump.
- Now this mechanism is used for test machine on produce vibrations.
- This is also used as a sine-cosine generator for computing elements.



Example 1: In a crank and slotted lever quick-return mechanism shown in Figure, the distance between the fixed centres is 300 mm and thee length of the driving crank is 150 mm. Find the inclination of the slotted lever with the vertical in the extreme position and the ratio of time of cutting stroke to return stroke.

Solution:

Given AC = 300 mm, B_1C = 150 mm

$$\cos(\beta/2) = B_1C / AC$$

$$= \frac{150}{300} = 0.50$$

$$\beta = 120°$$

$$\alpha = 360° - 120° = 240°$$

$$\frac{Time \quad of \quad cutting \quad stroke}{Time \quad of \quad return \quad stroke} = \frac{\alpha}{\beta} = \frac{240°}{120°} = 2$$

Inclination of slotted lever with the vertical $= 90° - \dfrac{\beta}{2} = 90° - \dfrac{120°}{2}$

$= 30°$

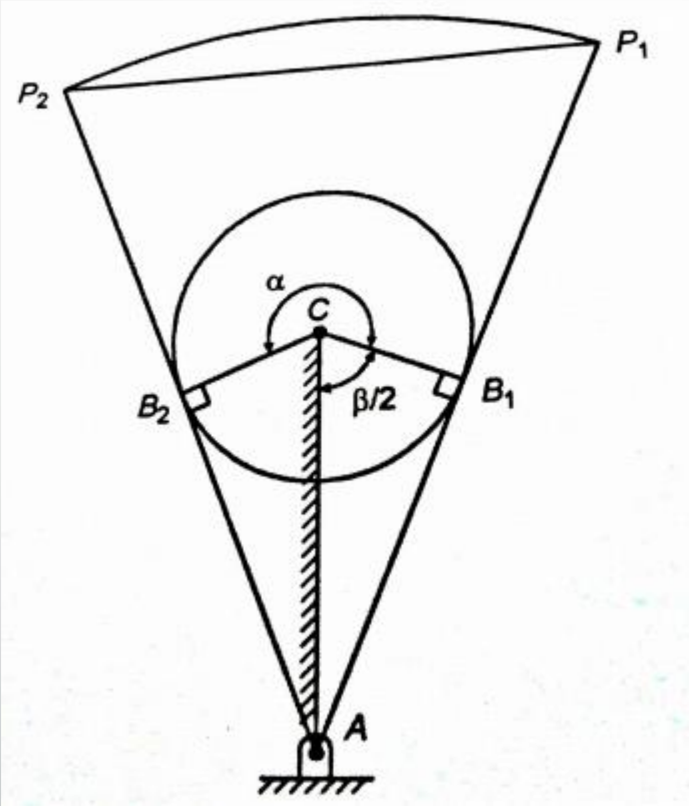


Fig. Crank and Slotted lever mechanism

Thank You